

META-ANALYSIS

Intracoronary Imaging-Guided Percutaneous Coronary Intervention

An Updated Meta-Analysis of Randomized Trials

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ABSTRACT

BACKGROUND Recent societal guidelines provide a Class I recommendation for intracoronary imaging (ICI)-guided percutaneous coronary intervention (PCI) for complex or left main coronary artery disease; however, recent trials have failed to show a benefit.

OBJECTIVES The study aim was to re-evaluate the efficacy of ICI-guided PCI and to explore heterogeneity of results across trials.

METHODS PubMed, Embase, Cochrane Central, and ClinicalTrials.gov databases were searched through June 3, 2026, for randomized controlled trials (RCTs) comparing outcomes with ICI-guided PCI vs angiography-guided PCI in the drug-eluting stent era. Direct comparison meta-analyses, mixed treatment comparison meta-analyses, trial sequential analyses, fragility analysis, and Bayesian meta-analyses were performed. Meta-regression analysis was performed to evaluate heterogeneity using the following variables: stent length (surrogate for lesion length), stent diameter, difference in postdilatation (percentage) between the groups, publication year, and geographic location.

RESULTS A total of 28 RCTs met the inclusion criteria, comprising 24,634 patients (mean age 64.8 years; 75.9% male) with a mean follow-up of 22 months. ICI-guided PCI reduced target lesion failure (relative risk [RR]: 0.72; 95% CI: 0.61-0.85; $I^2 = 61\%$), cardiac mortality (RR: 0.73; 0.58-0.91; $I^2 = 5\%$), myocardial infarction (RR: 0.86; 0.75-0.97; $I^2 = 0\%$), target vessel revascularization (RR: 0.69; 95% CI: 0.57-0.83, $I^2 = 41\%$), stent thrombosis (RR: 0.55; 95% CI: 0.40-0.76; $I^2 = 0\%$), and all-cause mortality (RR: 0.82; 95% CI: 0.70-0.96; $I^2 = 0\%$) compared with angiography-guided PCI. Trial sequential analysis confirmed evidence for at least a 20% reduction in all outcomes with ICI-guided PCI. The fragility index was robust, ranging from 9 to 50. The posterior probability for benefit was >99% for all outcomes. Trial differences in geography and lesion characteristics explained most of the between-trial variance (residual tau = 0).

CONCLUSIONS In this updated meta-analysis of RCTs, after accounting for modifier effects from recent studies, ICI-guided PCI was associated with significantly reduced cardiovascular events compared with angiography-guided PCI. Geographic heterogeneity across trials was present mainly for target vessel revascularization, explained in large part by differences in lesion characteristics and procedural technique. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS
AND ACRONYMS**HTE** = heterogeneity of treatment effect**ICI** = intracoronary imaging**IVUS** = intravascular ultrasound**MI** = myocardial infarction**MSA** = minimal stent area**OCT** = optical coherence tomography**PCI** = percutaneous coronary intervention**RCT** = randomized clinical trial**RR** = relative risk**TLF** = target lesion failure**TSA** = trial sequential analysis**TVR** = target vessel revascularization

Both U.S. and European societal guidelines strongly recommend use of intracoronary imaging (ICI) (with either intravascular ultrasound [IVUS] or optical coherence tomography [OCT]) guidance during percutaneous coronary intervention (PCI), especially for complex coronary disease.^{1,2} These guidelines were based on trials and meta-analyses showing significant reduction in clinical outcomes with use of ICI-guided PCI.^{3,4} In a meta-analysis of ICI-guided PCI compared with angiography-guided PCI that included 22 trials published through August 2023, ICI-guided PCI was associated with a 29% reduction in the risk of target lesion failure (TLF) driven by reductions in the risks of cardiac death, target vessel-related myocardial infarction (MI), and target lesion revascularization.⁴ However, 2 recent European trials failed to show

superiority of IVUS-guided PCI compared with angiography-guided PCI, one in complex coronary lesions (IVUS-CHIP [Intravascular Ultrasound Guidance for Complex High-Risk Indicated Procedures]) and the other in unprotected left main coronary artery disease (OPTIMAL [Optimization of Left Main PCI With Intravascular Ultrasound]).^{5,6} In contrast, a third recent trial from Asia in patients with complex bifurcation lesions (DKCRUSH VIII [IVUS-guided DK Crush Stenting Technique for Patients With Complex Bifurcation Lesions]) found that IVUS-guided PCI improved 1-year outcomes compared with angiography-guided PCI.⁷ Several other large-scale randomized controlled trials (RCTs) have also been reported since the prior systematic reviews.^{8,9}

Taken together, these findings have re-ignited the debate and highlight the need to re-evaluate the role of ICI-guided PCI in contemporary practice. Accordingly, the objective of this updated systematic review and meta-analysis was to compare the efficacy of ICI-guided PCI (with IVUS or OCT) vs angiography-

guided PCI and to evaluate heterogeneity of the results across trials.

METHODS

We conducted a systematic review and meta-analysis in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.¹⁰ The data that support the findings of this study are available either in this article or in the [Supplemental Appendix](#). Because this was a trial-level meta-analysis that used published data, it did not meet criteria for Institutional Board Review.

SEARCH STRATEGY, SELECTION CRITERIA, AND STUDY OUTCOMES. We searched relevant trials using PubMed, Embase, Cochrane Central, and ClinicalTrials.gov databases through June 3, 2026. The following key words were used for this search: “intra-coronary imaging,” “intravascular ultrasound,” “optical coherence tomography,” “angiography,” and “percutaneous coronary interventions” ([Supplemental Table 1](#)). The search strategy was developed by using Boolean operators, including “AND” and “OR,” to combine relevant key words, subject headings, and their synonyms across the aforementioned databases. Further studies were identified using the references of the eligible studies. There were no language or publication restrictions for the search strategy.

The inclusion criteria were RCTs that compared the efficacy of ICI-guided PCI vs angiography-guided PCI in the drug-eluting stent era. ICI-guided trials using bare-metal stents, reviews, observational studies, case reports, letters, editorials, and studies without a comparator arm were excluded.

PICO (Population, Intervention, Comparison, and Outcome) criteria were used for study selection.¹¹ The population of interest was patients undergoing PCI; intervention was ICI-guided PCI (IVUS or OCT); comparison was angiography-guided PCI; and outcomes were TLF, cardiac mortality, MI, TVR, definite

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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or probable stent thrombosis, and all-cause mortality. The definition of TLF varied across trials (Supplemental Table 2).

DATA EXTRACTION AND RISK OF BIAS ASSESSMENT.

Two authors independently screened the studies (M.H.M. and R.S.Z.), and full-text studies were then screened for final selection based on the aforementioned and predetermined inclusion criteria. These 2 authors independently reviewed the literature, extracted and entered data, and appraised the risk of bias of included trials using the Cochrane Risk of Bias Tool. This tool was used to assess study randomization process, deviation from intended intervention, missing outcome data, measurement of the outcome, selection of the reported results, and overall bias (Supplemental Table 3).^{12,13} Any discrepancies were resolved by arbitration with the first author (S.B.).

STATISTICAL ANALYSIS. Continuous variables are reported as mean $\pm$ SD, and categorical variables are reported as frequencies. The direct meta-analysis compared ICI-guided PCI (with either IVUS or OCT) with angiography-guided PCI. Relative risks (RRs) and 95% CIs were calculated for categorical variables by using random-effects models with the DerSimonian-Laird method.¹⁴ Random effects were chosen for conservative interpretation. For sparse event data, a continuity correction was applied by adding 0.5 to each cell of studies containing zero-event counts. Heterogeneity between studies was assessed by using the I^2 statistic with <25% indicating low, 25% to 50% moderate, 50% to 75% high, and >75% extreme heterogeneity, respectively.¹⁵ Publication bias was assessed by using funnel plots and Egger regression tests. The fragility index, defined as the minimum number of event changes required to alter statistical significance, was calculated for pooled RRs by using a random-effects model.¹⁶ A fragility index <5 was considered to be fragile and was computed iteratively by using the metafrag command in Stata (StataCorp).

To address the risk of type I and type II errors inherent in repeated testing and cumulative meta-analyses, a trial sequential analysis (TSA) was performed. Analogous to interim analyses in RCTs, TSA applies monitoring boundaries, based on DeMets stopping rules, to determine whether the evidence is sufficient to confirm an anticipated effect or indicate futility. TSA (version 0.9 Beta¹⁷) was conducted by using established methodology,^{18,19} assuming a 20% relative decrease in outcomes (chosen to match the nominal effect seen in most cardiovascular trials that are both clinically meaningful and realistic), with $\alpha = 5\%$ and power = 80%. The resulting trial

TABLE 1 Summary of Direct Comparison Meta-Analyses

Endpoints	Direct Comparison Meta-Analysis		
	Relative Risk (95% CI)	Heterogeneity (I^2)	Fragility Index
Target lesion failure	0.72 (0.61-0.85)	61%	50
Cardiac mortality	0.73 (0.58-0.91)	5%	10
Myocardial infarction	0.86 (0.75-0.97)	0%	9
Target vessel revascularization	0.69 (0.57-0.83)	41%	36
Definite or probable stent thrombosis	0.55 (0.40-0.76)	0%	15
All-cause mortality	0.82 (0.70-0.96)	0%	11

sequential monitoring boundaries indicate whether additional evidence is needed (boundary not crossed) or whether firm evidence for effect or futility has been reached (boundary crossed).

Subgroup analyses were conducted based on: 1) trial era (2020 or earlier vs later trials); and 2) trial geography (predominately Asian vs predominately non-Asian trials). Meta-regression analysis was also performed to explore heterogeneity of results across the trials. The variables included: stent length (a proxy for lesion length), stent diameter in angiography arm (proxy for vessel size), percentage of postdilatation difference between ICI-guided PCI vs angiography-guided PCI (proxy for stent optimization), year of publication, and geographic trial location.

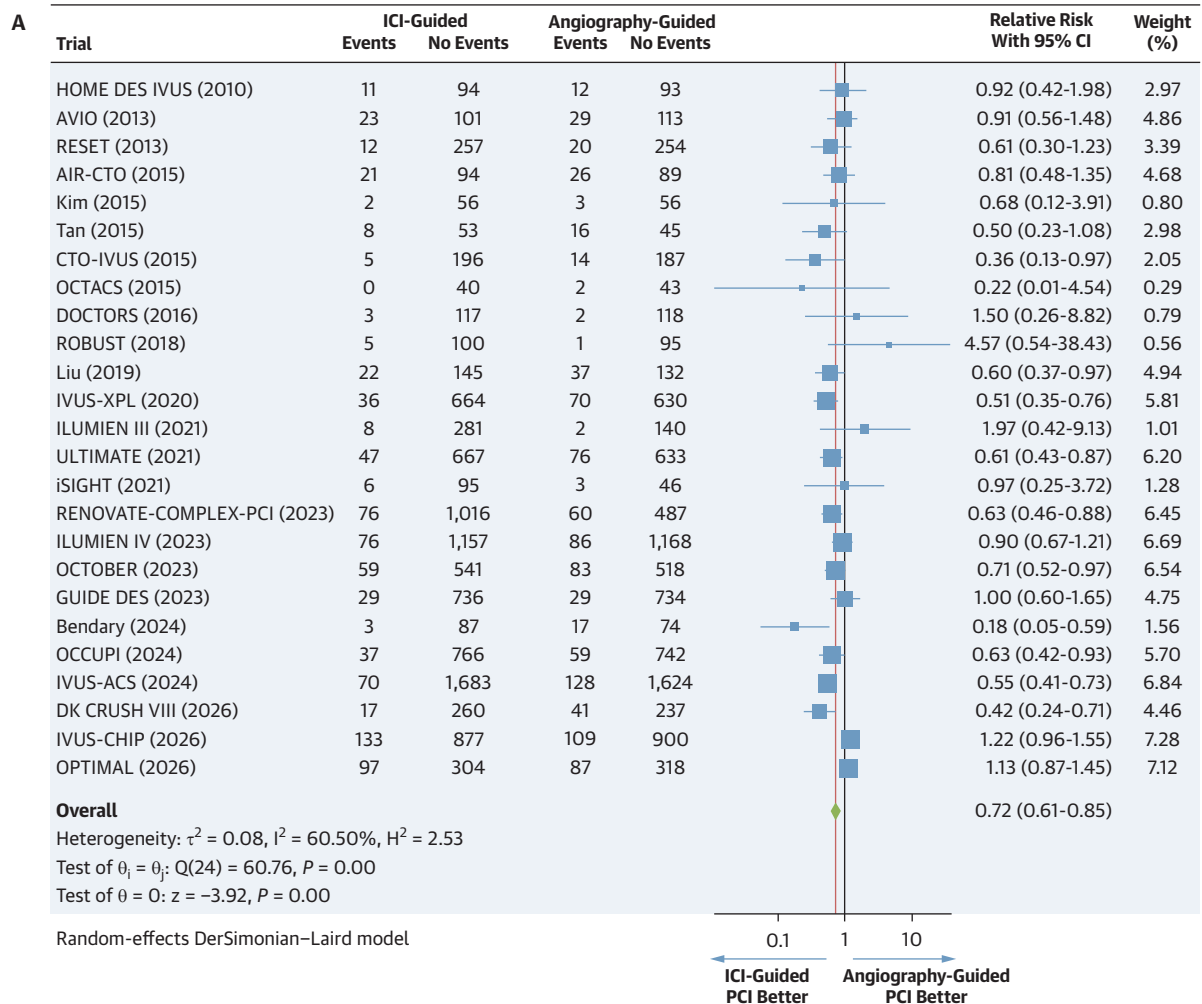
We also performed a mixed treatment network meta-analysis comparing IVUS-guided PCI and OCT-guided PCI with angiography-guided PCI. Effect size was evaluated with a random-effects model with angiography-guided PCI as the reference treatment arm (Supplemental Methods and Results).²⁰⁻²³ Finally, Bayesian random-effects meta-analyses were performed as supportive analyses for each endpoint.²⁴ Posterior mean and posterior median effect estimates with 95% credible intervals were calculated. Posterior probabilities were reported as the probability that the RR was <1.00 and <0.90, representing any reduction and at least a 10% relative reduction, respectively.

All analyses were performed in the intention-to-treat population, and a 2-sided P value <0.05 was considered significant. Analyses were conducted by using Stata version 19.5 software (StataCorp).

RESULTS

The initial literature search yielded 650 reports, from which 28 RCTs were included in the final analysis (Supplemental Figure 1).^{5-9,25-47}

BASELINE CHARACTERISTICS. Of the 28 RCTs, 17 trials compared IVUS-guided PCI vs angiography-

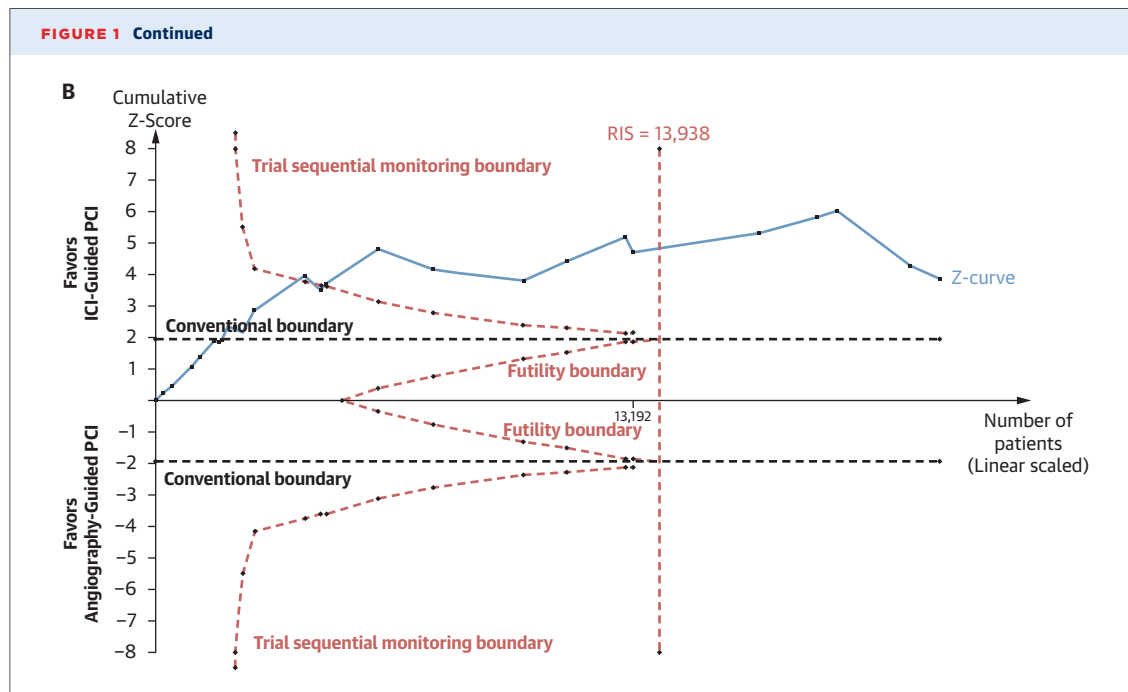
FIGURE 1 Target Lesion Failure

(A) Direct comparison meta-analysis of intracoronary imaging (ICI)-guided percutaneous coronary intervention (PCI) compared with angiography-guided PCI in target lesion failure. (B) Trial sequential analysis evaluating for a 20% relative reduction with ICI-guided PCI with $\alpha = 5\%$ and power = 80%. RIS = required information size.

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guided PCI,^{5-7,9,25,26,31,34-37,40,41,43-45,47} 7 trials compared OCT-guided PCI vs angiography-guided PCI,^{8,27,32,33,38,39,42} 3 trials compared OCT-guided PCI vs IVUS-guided PCI,²⁸⁻³⁰ and 1 trial compared IVUS- or OCT-guided PCI vs angiography-guided PCI.⁴⁶ Overall, 24,634 patients were enrolled with mean age of 64.8 ± 2.8 years; 75.9% of randomized participants were male. Mean duration of follow-up was 21.6 ± 12.4 months (Supplemental Table 2). Domain level and overall risk of bias are reported in Supplemental Table 3. All trials were judged to be low risk of bias.

FREQUENTIST META-ANALYSIS. Target lesion failure. A total of 25 trials evaluated TLF between ICI-guided PCI and angiography-guided PCI, with 806 (7.2%) of 11,193 and 1,012 (9.7%) of 10,488 events, respectively. The risk of TLF was lower with ICI-guided PCI compared with angiography-guided PCI (RR: 0.72; 95% CI: 0.61-0.85; fragility index = 50) (Figure 1A, Table 1). There was high heterogeneity ($I^2 = 61\%$) in this analysis but no evidence of publication bias (Supplemental Figure 2A). In the TSA, the cumulative Z-curve crossed both the conventional boundary and the trial sequential



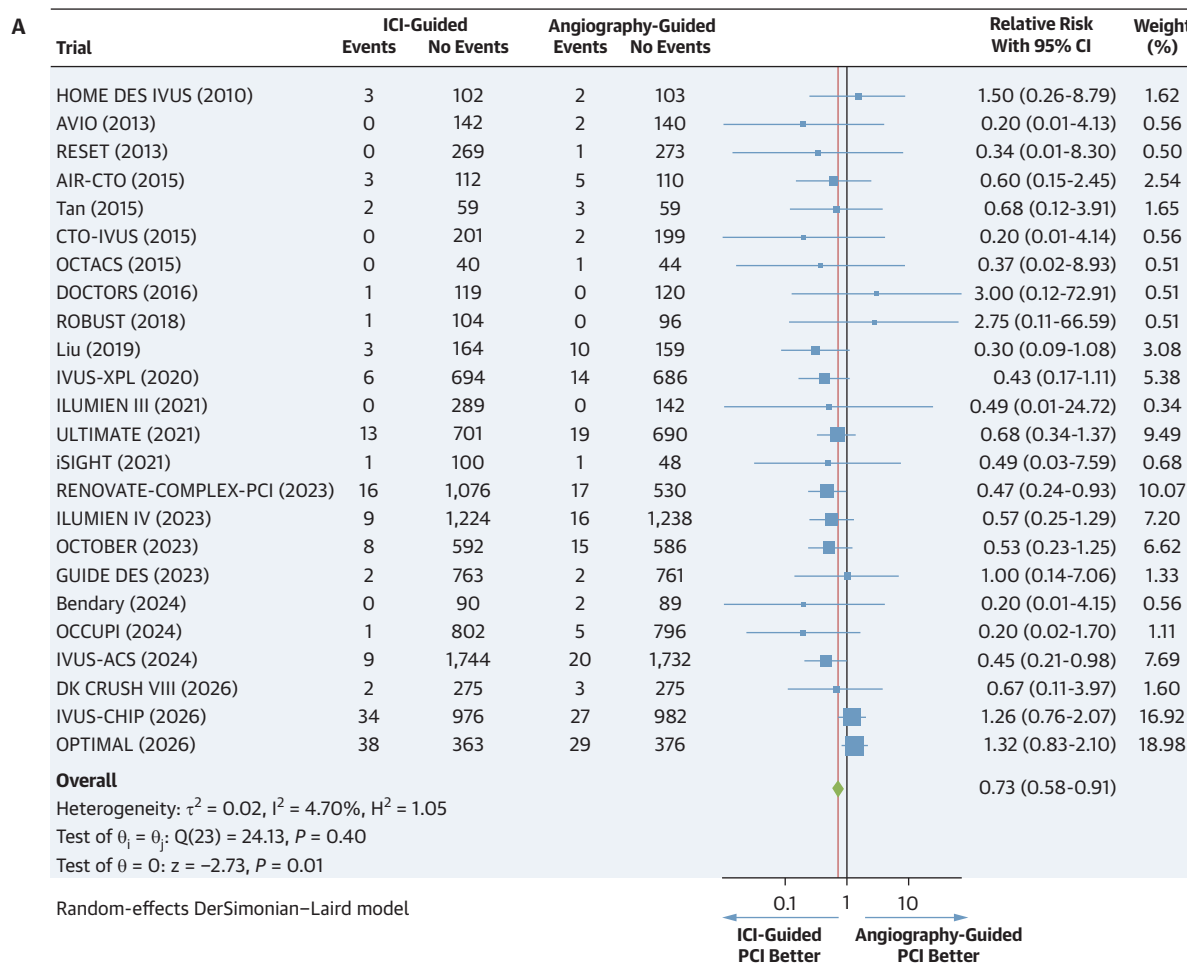
monitoring boundary, indicating that there was firm evidence for at least a 20% reduction in TLF with ICI-guided PCI compared with angiography-guided PCI (**Figure 1B**).

Subgroup analysis based on trial era (2020 or earlier vs later trials) showed no heterogeneity of treatment effect (HTE) ($P_{\text{interaction}} = 0.43$) (**Supplemental Figure 2B**). However, subgroup analysis based on trial geography showed significant HTE ($P_{\text{interaction}} < 0.001$) with lower risk with ICI-guided PCI compared with angiography-guided PCI in trials conducted in Asia (RR: 0.60; 95% CI: 0.53-0.68; $I^2 = 0\%$) but not elsewhere (RR: 0.95; 95% CI: 0.76-1.18; $I^2 = 46\%$) (**Supplemental Figure 2C**). Similarly, meta-regression analysis (unadjusted) showed that the difference between ICI-guided PCI vs angiography-guided PCI correlated significantly ($P < 0.001$) with the trial's geographic location (**Supplemental Figure 2D**). However, in a meta-regression model that included all of the prespecified variables (including differences in lesion characteristics and technique) (model $P = 0.041$), the trial's geographic location no longer modified the effect of ICI-guided PCI vs angiography-guided PCI ($P = 0.19$). Differences in lesion/stent characteristics explained most of the trial variance by geography (residual tau = 0) (**Supplemental Table 4**).

Cardiac mortality. A total of 24 trials evaluated cardiac mortality between ICI-guided PCI and angiography-guided PCI, with 152 (1.4%) of 11,153 and

196 (1.9%) of 10,430 events, respectively. There was a lower risk of cardiac mortality with ICI-guided PCI compared with angiography-guided PCI (RR: 0.73; 95% CI: 0.58-0.91; fragility index = 10) (**Figure 2A**, **Table 1**). There was low heterogeneity ($I^2 = 5\%$) and no evidence of publication bias (**Supplemental Figure 3A**). In the TSA, the cumulative Z-curve crossed both the conventional boundary and the trial sequential monitoring boundary, indicating that there was firm evidence for at least a 20% reduction in cardiac mortality with ICI-guided PCI compared with angiography-guided PCI (**Figure 2B**).

Subgroup analysis based on trial era showed no significant HTE ($P_{\text{interaction}} = 0.29$) (**Supplemental Figure 3B**). In contrast, subgroup analysis based on geographic location showed significant HTE ($P_{\text{interaction}} < 0.001$) with lower risk of ICI-guided PCI compared with angiography-guided PCI in trials conducted in Asia (RR: 0.50; 95% CI: 0.36-0.69; $I^2 = 0\%$) but not elsewhere (RR: 1.02; 95% CI: 0.77-1.35; $I^2 = 0\%$) (**Supplemental Figure 3C**). Similarly, meta-regression analysis (unadjusted) showed that the difference between ICI-guided PCI vs angiography-guided PCI was greater in trials conducted in Asia ($P = 0.004$), in those with shorter stent length ($P = 0.008$), and in those with smaller stent diameter in the angiography-guided arm ($P = 0.005$) (**Supplemental Figure 3D**). However, in a meta-regression model including all of the prespecified variables, no individual covariate (including

FIGURE 2 Cardiac Mortality

(A) Direct comparison meta-analysis of ICI-guided PCI compared with angiography-guided PCI in cardiac mortality. (B) Trial sequential analysis evaluating for a 20% relative reduction with ICI-guided PCI with $\alpha = 5\%$ and power = 80%. Abbreviations as in [Figure 1](#).

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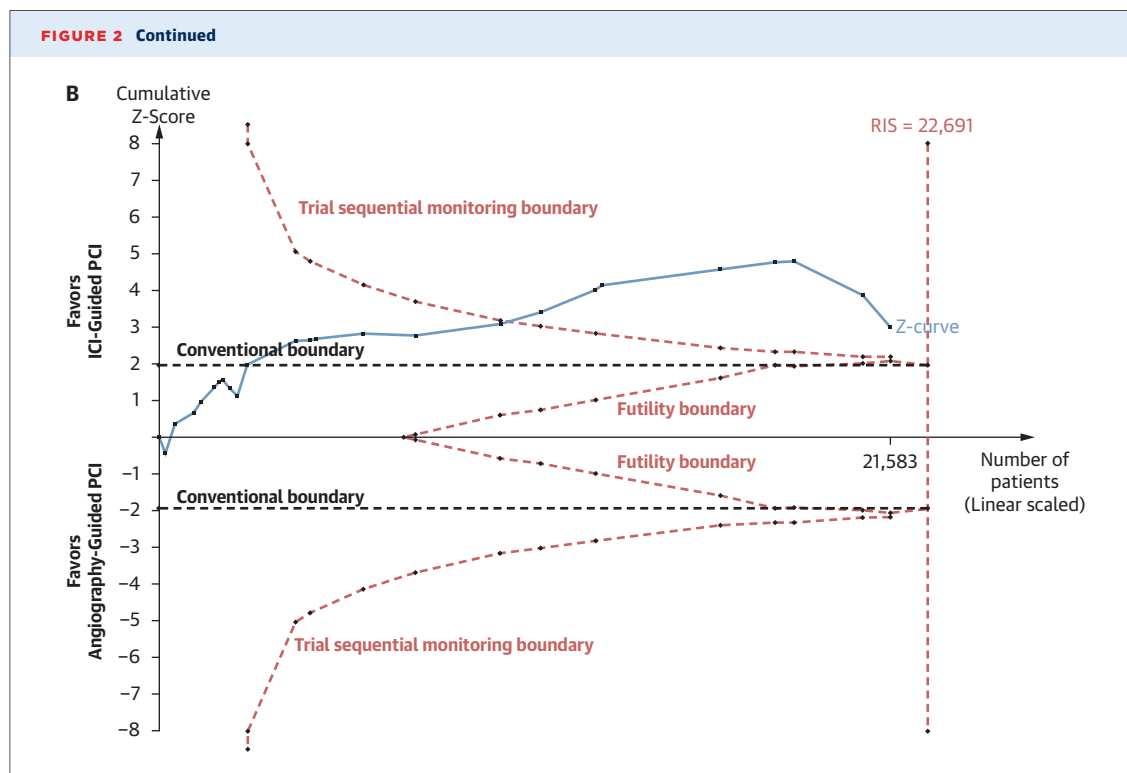
geography; $P = 0.93$) significantly modified the effect of ICI-guided PCI vs angiography-guided PCI ([Supplemental Table 4](#)).

Myocardial infarction. A total of 23 trials evaluated MI between ICI-guided PCI and angiography-guided PCI, with 413 (4.39%) of 9,400 and 455 (5.24%) of 8,678 events, respectively. There was a lower risk of MI with ICI-guided PCI compared with angiography-guided PCI (RR: 0.86; 95% CI: 0.75-0.97; fragility index = 9) ([Figure 3A](#), [Table 1](#)). There was no heterogeneity ($I^2 = 0\%$) in the analysis and no evidence of publication bias ([Supplemental Figure 4A](#)). In the TSA, the cumulative Z-curve crossed both the conventional boundary and the trial sequential monitoring

boundary, indicating that there was firm evidence for at least a 20% reduction in MI with ICI-guided PCI compared with angiography-guided PCI ([Figure 3B](#)).

Subgroup analysis showed no significant HTE of ICI-guided PCI vs angiography-guided PCI on MI based on trial era ($P_{\text{interaction}} = 0.69$) or trial geography ($P_{\text{interaction}} = 0.09$) ([Supplemental Figures 4B and 4C](#)). Similarly, meta-regression analysis showed no significant relationships between any evaluated covariate and the effect of ICI-guided PCI vs angiography-guided PCI ([Supplemental Figure 4D](#), [Supplemental Table 4](#)).

Target vessel revascularization. A total of 24 trials evaluated TVR between ICI-guided PCI and



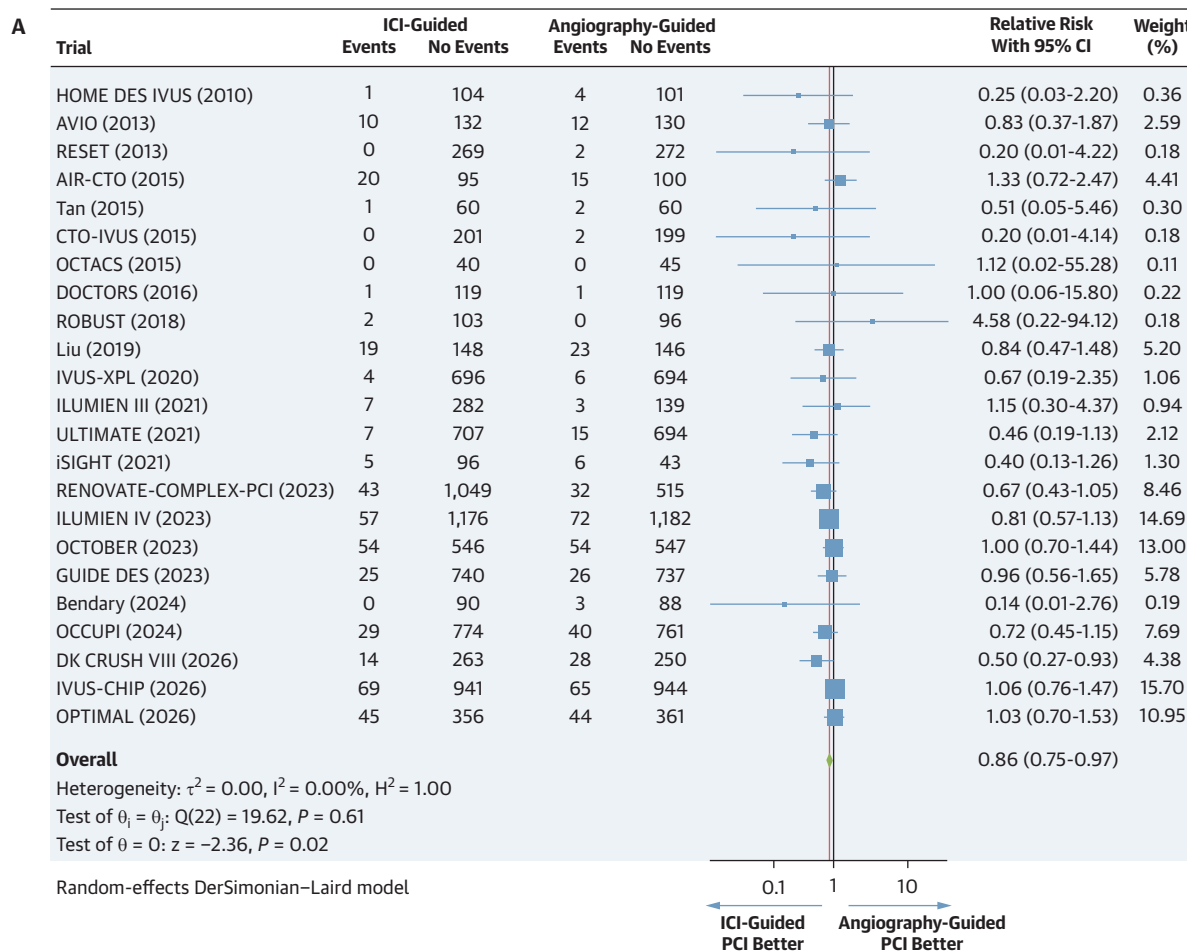
angiography-guided PCI, with 435 (3.89%) of 11,171 and 567 (5.43%) of 10,444 events, respectively. There was a lower risk of TVR with ICI-guided PCI compared with angiography-guided PCI (RR: 0.69; 95% CI: 0.57-0.83; fragility index = 36) (**Figure 4A**, **Table 1**). There was moderate heterogeneity ($I^2 = 41\%$) in the analysis and no evidence of publication bias (**Supplemental Figure 5A**). In the TSA, the cumulative Z-curve crossed both the conventional boundary and the trial sequential monitoring boundary, indicating that there was firm evidence for at least a 20% reduction in TVR with ICI-guided PCI compared with angiography-guided PCI (**Figure 4B**).

Subgroup analysis based on trial era showed similar effect without significant HTE ($P_{\text{interaction}} = 0.35$) (**Supplemental Figure 5B**). In contrast, subgroup analysis based on geographic location showed significant HTE ($P_{\text{interaction}} < 0.001$) with lower risk of ICI-guided PCI compared with angiography-guided PCI in trials conducted in Asia (RR: 0.55; 95% CI: 0.46-0.66; $I^2 = 0\%$) but not elsewhere (RR: 0.97; 95% CI: 0.82-1.15; $I^2 = 0\%$) (**Supplemental Figure 5C**). Similarly, meta-regression analysis (unadjusted) showed that the difference between ICI-guided PCI vs angiography-guided PCI significantly correlated with trial's geographic location ($P < 0.001$) (**Supplemental Figure 5D**). However,

in a meta-regression model including all of the pre-specified variables, geographic location no longer modified the effect of ICI-guided PCI vs angiography-guided PCI ($P = 0.15$) (**Supplemental Table 4**).

Definite or probable stent thrombosis. A total of 25 trials evaluated definite or probable stent thrombosis between ICI-guided PCI and angiography-guided PCI, with 60 (0.54%) of 11,212 and 115 (1.10%) of 10,489 events, respectively. There was a lower risk of definite or probable stent thrombosis with ICI-guided PCI compared with angiography-guided PCI (RR: 0.55; 95% CI: 0.40-0.76; fragility index = 15) (**Figure 5A**, **Table 1**). There was no heterogeneity ($I^2 = 0\%$) in the analysis and no evidence of publication bias (**Supplemental Figure 6A**). In the TSA, the cumulative Z-curve crossed both the conventional boundary and the trial sequential monitoring boundary, indicating that there was firm evidence for at least a 20% reduction in stent thrombosis with ICI-guided PCI compared with angiography-guided PCI (**Figure 5B**).

Subgroup analysis showed no significant HTE based on trial era ($P_{\text{interaction}} = 0.96$) (**Supplemental Figure 6B**) or trial geography ($P_{\text{interaction}} = 0.42$) (**Supplemental Figure 6C**). Similarly, meta-regression analysis revealed no significant relationships between any evaluated covariate and the effect of

FIGURE 3 Myocardial Infarction

(A) Direct comparison meta-analysis of ICI-guided PCI compared with angiography-guided PCI in myocardial infarction. (B) Trial sequential analysis evaluating for a 20% relative reduction with ICI-guided PCI with $\alpha = 5\%$ and power = 80%. Abbreviations as in [Figure 1](#).

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ICI-guided PCI vs angiography-guided PCI ([Supplemental Figure 6D](#), [Supplemental Table 4](#)).

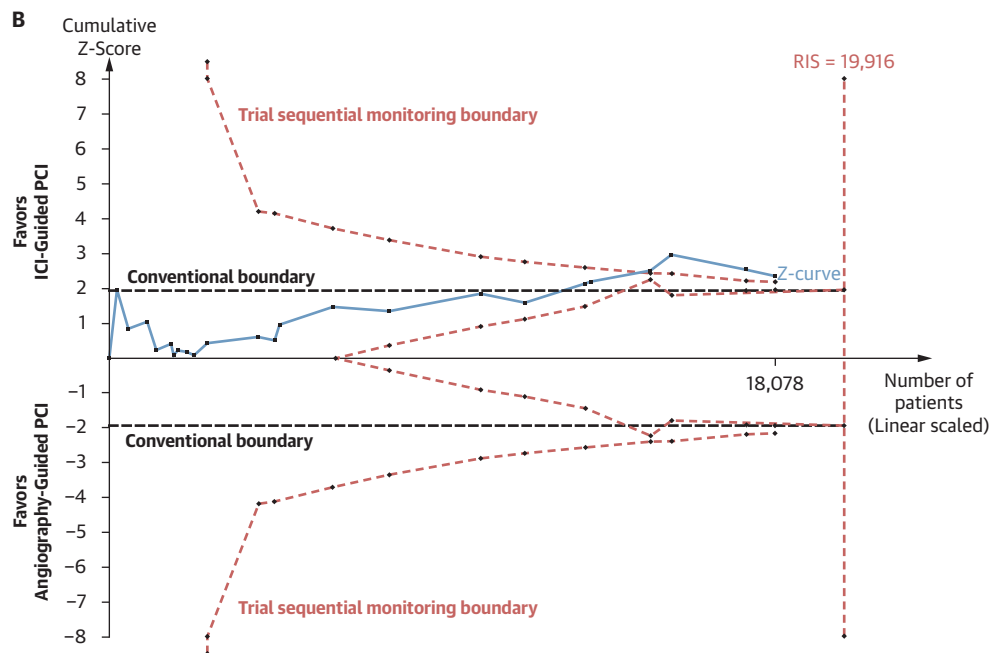
All-cause mortality. A total of 23 trials evaluated all-cause mortality between ICI-guided PCI and angiography-guided PCI, with 289 (2.61%) of 11,063 and 332 (3.21%) of 10,339 deaths, respectively. There was a lower risk of all-cause mortality with ICI-guided PCI compared with angiography-guided PCI (RR: 0.82; 95% CI: 0.70-0.96; fragility index = 11) ([Figure 6A](#), [Table 1](#)). There was no heterogeneity ($I^2 = 0\%$) in the analysis and no evidence of publication bias ([Supplemental Figure 7A](#)). In the TSA, the cumulative Z-curve crossed both the conventional boundary and the trial sequential monitoring boundary, indicating that there was firm evidence for

at least a 20% reduction in all-cause mortality with ICI-guided PCI compared with angiography-guided PCI ([Figure 6B](#)).

Subgroup analysis showed no significant HTE based on trial era ($P_{\text{interaction}} = 0.31$) ([Supplemental Figure 7B](#)) or trial geography ($P_{\text{interaction}} = 0.34$) ([Supplemental Figure 7C](#)). Meta-regression analysis (unadjusted) showed that the difference between ICI-guided PCI vs angiography-guided PCI correlated with stent length ($P = 0.048$) ([Supplemental Figure 7D](#)). However, this relationship was no longer significant ($P = 0.34$) in a model including all covariates ([Supplemental Table 4](#)).

MIXED TREATMENT COMPARISON META-ANALYSIS. The mixed treatment network meta-analysis showed that

FIGURE 3 Continued



the effects of ICI-guided PCI were similar with both IVUS-guided PCI and OCT-guided PCI compared with angiography-guided PCI (Supplemental Figures 2E, 3E, 4E, 5E, 6E, 7E, and 8, Supplemental Table 5).

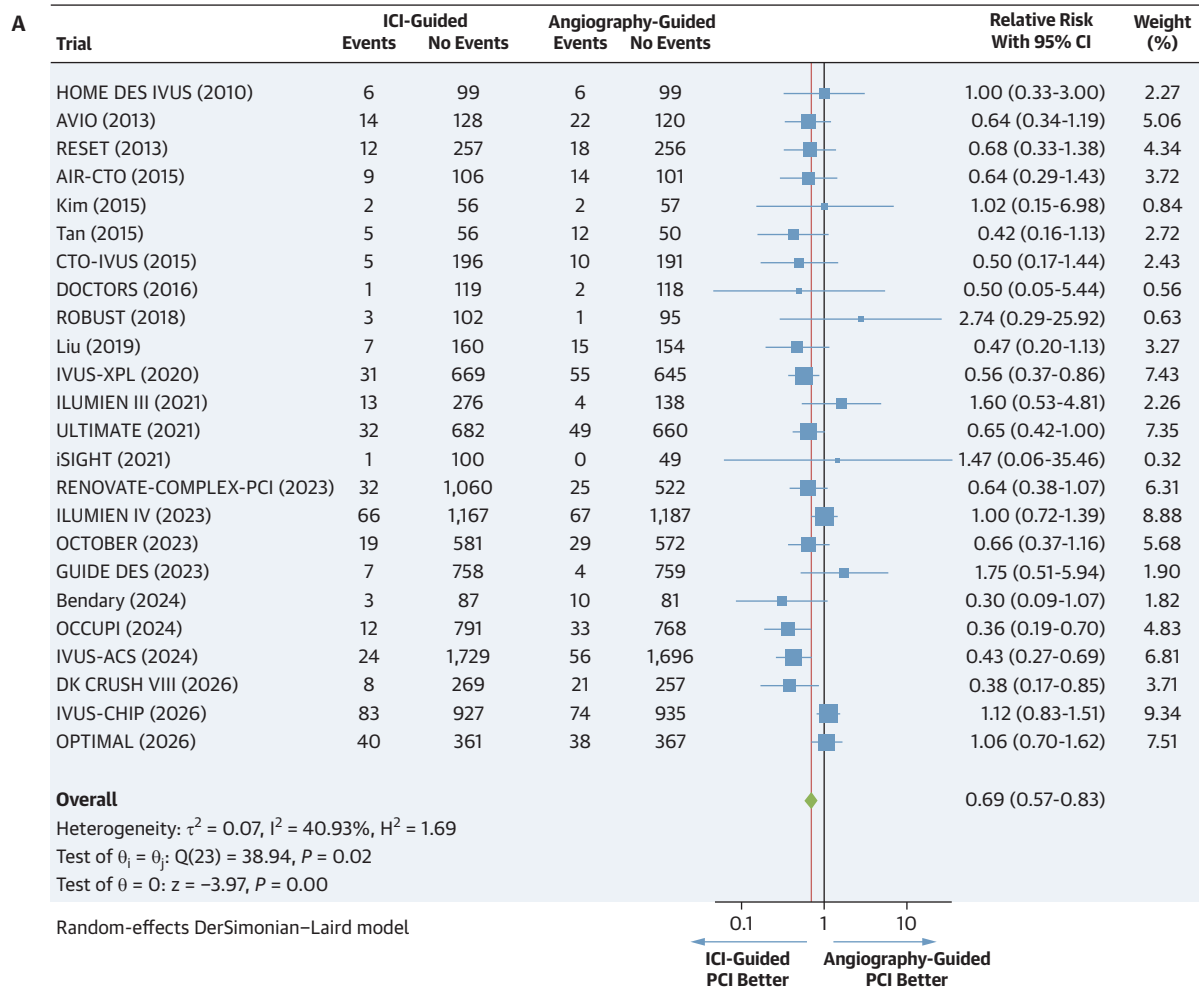
BAYESIAN META-ANALYSIS. Bayesian analyses were concordant with the frequentist results. The posterior probability of any benefit with ICI-guided PCI in all trials was >99% for all outcomes (Supplemental Figure 9, Supplemental Table 6) and in Asian trials was ≥98% (Supplemental Table 6). For trials conducted outside Asia, the posterior probability for benefit of ICI-guide PCI ranged from 70% for TLF to 95% for stent thrombosis.

DISCUSSION

This systematic review and meta-analysis of 28 RCTs in which 24,634 patients were randomized to treatment provides a comprehensive and contemporary evaluation of ICI-guided PCI compared with angiography-guided PCI, particularly in the context of recently published trials that have elicited some controversy. Our findings show that ICI-guided PCI results in consistent and clinically meaningful reductions across multiple endpoints, including TLF, MI, TVR, stent thrombosis, and both cardiac and all-cause mortality. Firm evidence for at least 20% reductions in all major outcomes was present despite

the inclusion of several recent trials with neutral results, suggesting that the totality of evidence continues to support ICI-guided intervention. The results were not fragile, with the fragility index varying from 9 to 50, indicating robustness of the findings. Bayesian meta-analysis confirmed these findings with a posterior probability for benefit with ICI-guided PCI of >99% for all outcomes. High heterogeneity was present for TLF, moderate heterogeneity was present for TVR, and no or low heterogeneity was present for all other outcomes. Trial differences in geography and lesion characteristics explained most of the between-trial variance. However, the subgroup and meta-regression analyses were exploratory. Finally, network meta-analysis found similar benefits with IVUS-guided PCI and OCT-guided PCI compared with angiography-guided PCI, with no major differences between IVUS-guided and OCT-guided PCI for any outcome.

ICI-guided PCI has had a winding path to acceptance. Early limitations, including increased procedural time, workflow complexity, and greater cost without reimbursement, contributed to relatively low utilization (approximately 15% of PCI procedures in the United States in 2019).⁴⁸ However, the accumulation of robust randomized trial data showing improvements in both lesion- and patient-level outcomes^{2,3} led to a rise in ICI-guided PCI in the United

FIGURE 4 Target Vessel Revascularization

(A) Direct comparison meta-analysis of ICI-guided PCI compared with angiography-guided PCI in target vessel revascularization. (B) Trial sequential analysis evaluating for a 20% relative reduction with ICI-guided PCI with $\alpha = 5\%$ and power = 80%. Abbreviations as in Figure 1.

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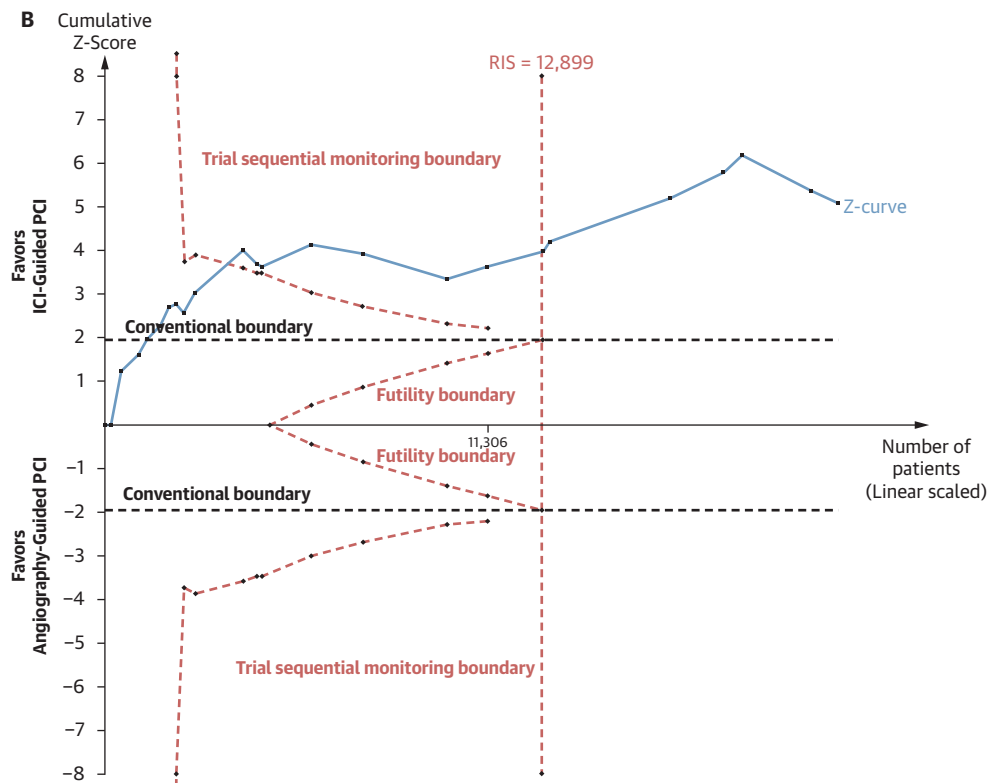
States to approximately 35% of cases by 2023,⁴³ followed by Class I guideline recommendations for ICI-guided PCI of complex coronary lesions in 2018, 2024, and 2025 in the Japanese, European, and U.S. societal guidelines.^{1,2,49}

Since the most recent published meta-analyses,^{3,4} there have been 6 new RCTs with 8,670 additional patients,^{5-9,26} representing a 53% increase in sample size beyond the prior evidence base. Although 4 of these trials^{7-9,26} reported significant improvements in outcomes with ICI-guided PCI, the IVUS-CHIP and OPTIMAL trials from Europe found no difference in most clinical outcomes between IVUS-guided PCI and angiography-guided PCI, re-introducing

uncertainty.^{5,6} The current updated meta-analysis was thus performed to re-examine the totality of the evidence and explore the heterogeneity of results across studies, specifically whether an underlying explanation might be found for the neutral results from these 2 new trials beyond chance.

The findings from the current study are consistent with those from an earlier meta-analysis (Supplemental Table 7). They reinforce the principal conclusion that ICI-guided PCI results in a significant and clinically meaningful reduction in stent-related adverse cardiovascular events (TLF), driven by 45%, 31%, and 27% reductions in stent thrombosis, TVR, and cardiac mortality, respectively, compared with

FIGURE 4 Continued

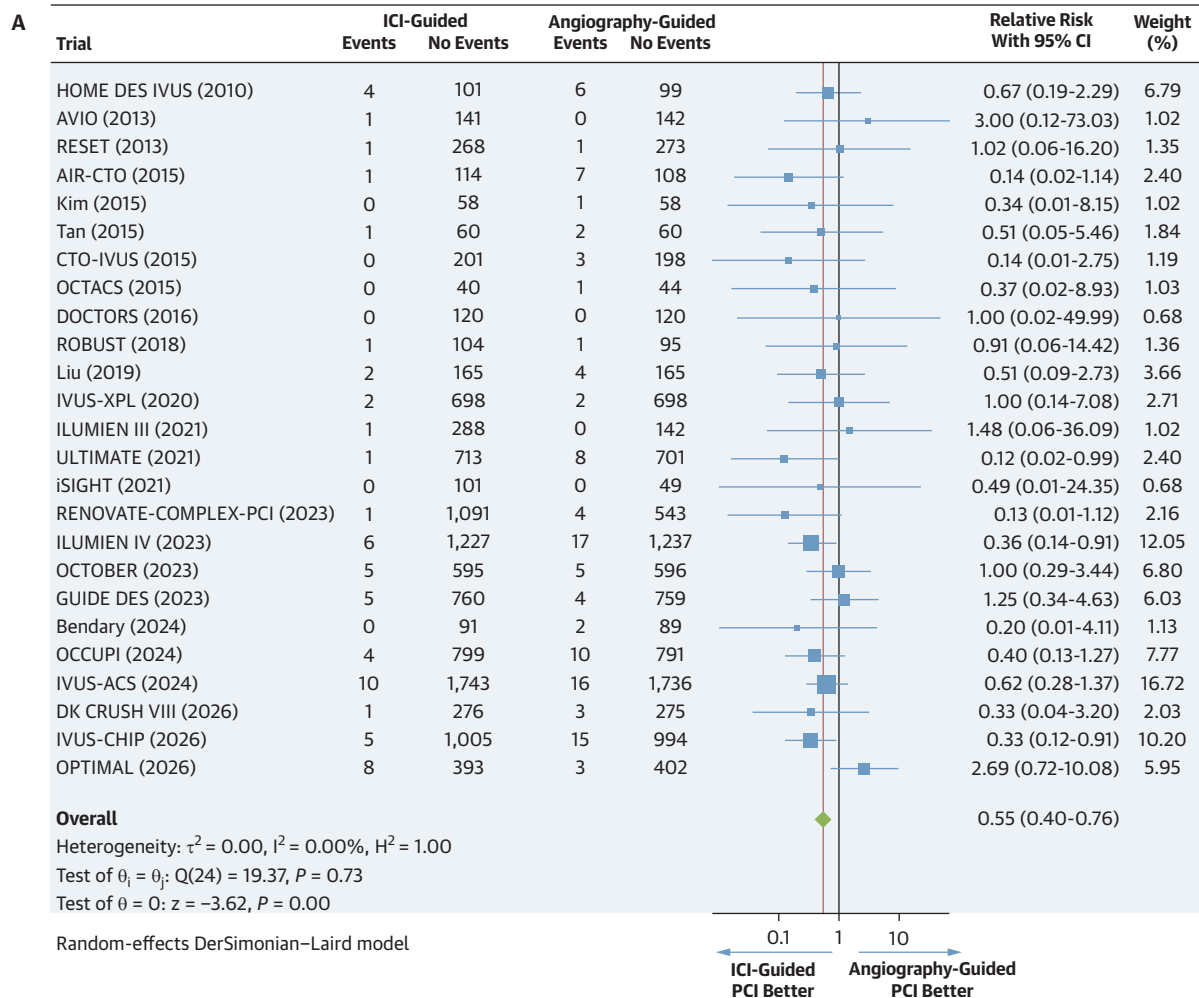


angiography-guided PCI. Even with the new data incorporated, none of these results was fragile, and the TSA results provide secure evidence for at least a 20% risk reduction in all endpoints with ICI guidance, with the Bayesian posterior probability of benefit of >99% for all outcomes.

The current analysis displayed high and moderate between-trial heterogeneity for the effects of ICI guidance vs angiography guidance on the outcomes of TLF and TVR, respectively, although little or no between-study heterogeneity was observed for all other outcomes. In unadjusted analysis, heterogeneity was observed in trials performed predominantly in Asian vs non-Asian countries for TLF, cardiac mortality, and TVR. However, in a multivariable meta-regression analysis incorporating trial-level lesion and procedural characteristics, geography was no longer an independent predictor of between-study variance for either TLF or TVR. Moreover, the lack of substantial between-study heterogeneity for stent thrombosis, MI, cardiac mortality, and all-cause mortality (as well as negative adjusted meta-regression findings) supports the interpretation that the positive effects of ICI

guidance on these outcomes in the overall meta-analysis extends to trials performed in Europe. Lastly, even in trials conducted outside Asia, the posterior probability for benefit of ICI-guided PCI according to Bayesian analysis ranged from 70% for TLF to 95% for stent thrombosis. However, these subgroup and meta-regression analyses of trial-level data are exploratory.

Although results of several trials of ICI-guided PCI from Europe have been positive,^{32,45} the recent neutral findings from the IVUS-CHIP and OPTIMAL trials^{5,6} have reintroduced questions as to the generalizability of the benefits of IVUS and OCT guidance. However, in the large-scale OCTOBER (European Trial on Optical Coherence Tomography Optimized Bifurcation Event Reduction) trial conducted in Europe, ICI-guided PCI significantly reduced major adverse cardiovascular events compared with angiography-guided PCI. Similarly, even in the IVUS-CHIP trial, there was a significant reduction in stent thrombosis with ICI-guided PCI compared with angiography-guided PCI.⁵ Several hypotheses have been offered to explain these apparently incongruent results. First, it has been

FIGURE 5 Definite or Probable Stent Thrombosis

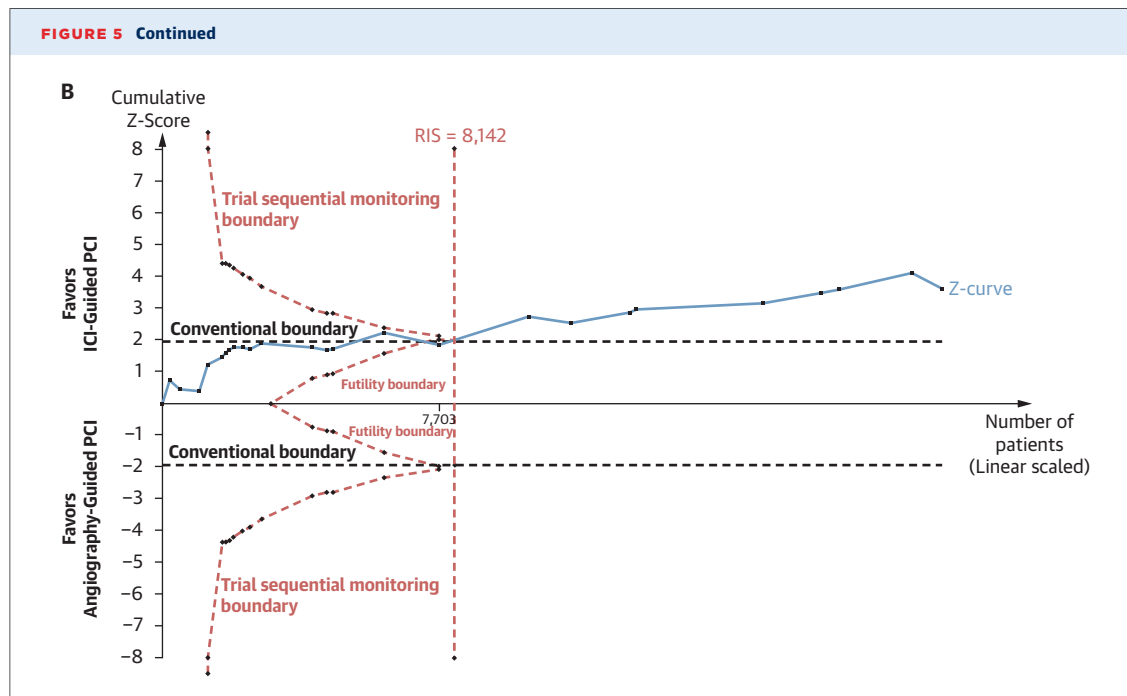
(A) Direct comparison meta-analysis of ICI-guided PCI compared with angiography-guided PCI in definite or probable stent thrombosis. (B) Trial sequential analysis evaluating for a 20% relative reduction with ICI-guided PCI with $\alpha = 5\%$ and power = 80%. Abbreviations as in Figure 1.

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suggested that the IVUS-CHIP and OPTIMAL operators may have translated learnings derived from their imaging experience to angiography-guided PCI, including routinely selecting larger and longer stents and postdilating. For example, postdilatation was used in only 6% more patients with ICI-guided PCI than with angiography-guided PCI in the European IVUS-CHIP trial (91% vs 85%, respectively), in contrast to 19% more with IVUS guidance in the Asian IVUS-XPL (Impact of Intravascular Ultrasound Guidance on the Outcomes of Xience Prime Stents in Long Lesions) trial (76% vs 57%).^{5,36} However, this theory is counterintuitive, as translating lessons from ICI

guidance to angiography guidance would most likely occur to a greater extent in Asian countries given their much greater utilization of ICI guidance than in Europe. In this regard, some have suggested that widespread reliance on ICI guidance in Asia can lead to suboptimal results in the angiography-guided group since angiography alone has become the less commonly used approach. However, although the average weighted stent length was longer with ICI guidance compared with the angiography guidance in non-Asian trials (44.9 mm vs 42.5 mm), as expected, the average weighted stent length was surprisingly shorter in the ICI group compared with the

FIGURE 5 Continued



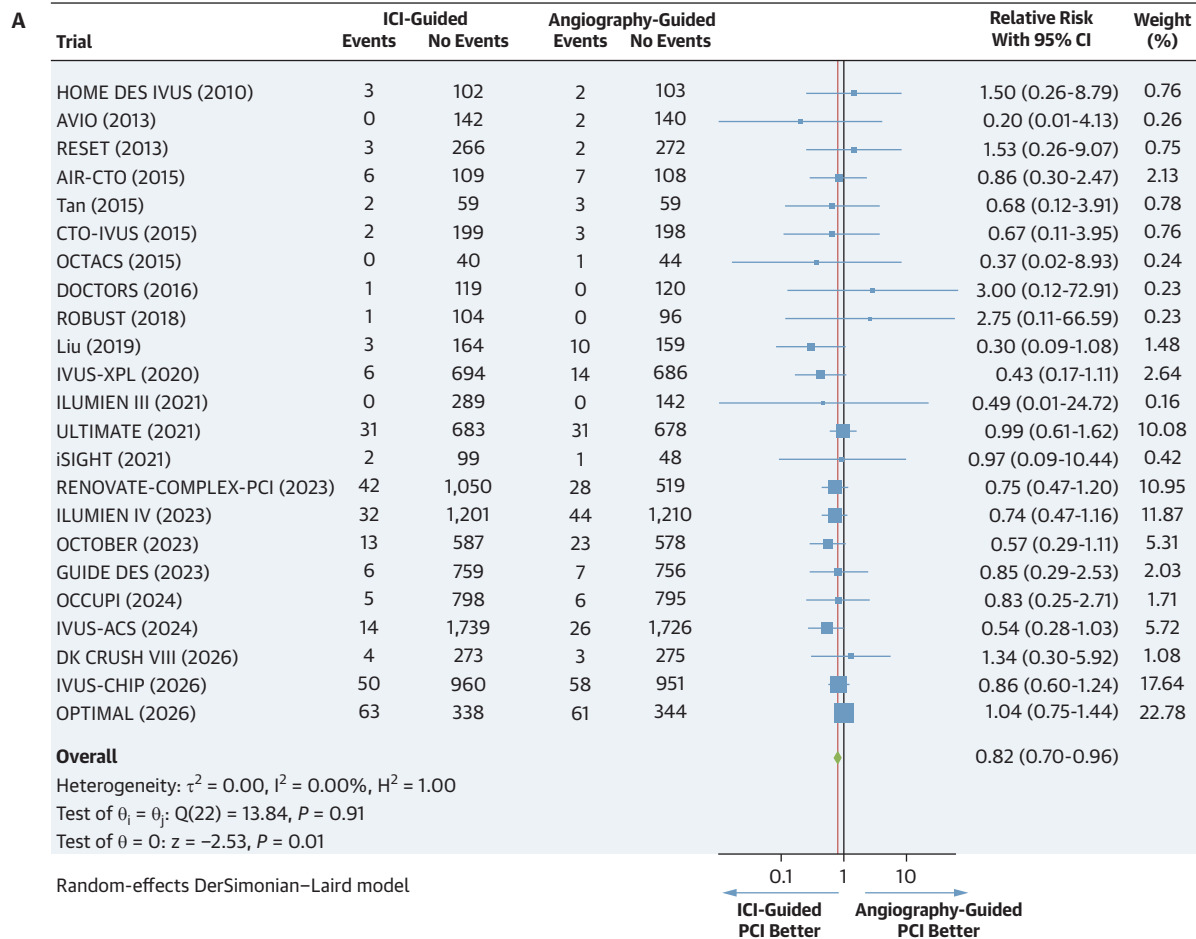
angiography-guided group in Asian trials (39.9 mm vs 43.5 mm), contradicting this theory. Unfortunately, most procedural details (and their rationale) that might affect outcomes are difficult to extract from these studies, leaving these theories speculative at best.

Second, within the ICI-guided arms of the recent European trials, the imaging findings may not have been optimally applied to maximize the minimal stent area (MSA) and treat edge issues, the principal determinants of restenosis and stent thrombosis.^{5,50} For example, in IVUS-CHIP, all predefined stent optimization criteria were achieved in only 48.0% of lesions; specifically, MSA $>5 \text{ mm}^2$ or $>90\%$ of the distal reference lumen was not met in 6.6% of lesions, residual plaque burden $\geq 50\%$ at the stent edges was present in 59.8% of lesions, and 2.9% of lesions had an edge dissection $>3 \text{ mm}$ in length.⁵ The extent to which the high rate of untreated residual peri-stent disease reflects the challenge in identifying an optimal landing zone in vessels with diffuse disease is unknown. In OPTIMAL, the target optimal MSA for the left main was set at 8.0 mm^2 , the standard used for vessels from Asia,⁵⁰ despite the fact that arteries in patients in Europe and the United States are substantially larger, requiring larger stent areas for adequate expansion.^{51,52} This likely explains why IVUS-driven stent optimization maneuvers were performed in fewer than one-third (29.3%) of patients.^{5,6} Achieving a larger MSA closer to the

external elastic lamina with IVUS-guided stent sizing may have further improved outcomes compared with lumen-based stent sizing with angiography guidance alone.

Third, a key limitation of the recent trials was the absence of systematic pre-PCI imaging, which many experts strongly advocate for to achieve accurate determination of vessel size, characterization of plaque and calcium distribution, guidance of lesion preparation strategies, and landing zone selection to avoid geographic miss.⁵³ In IVUS-CHIP, pre-PCI IVUS was recommended but not required and was performed in 63.7% of cases.⁵ Similarly, in the OPTIMAL trial of left main intervention, pre-PCI IVUS was performed in only 65.1% of lesions.⁶ In contrast, in the DKCRUSH VIII trial of bifurcation stenting (including left main intervention), pre-PCI IVUS was performed in $>95\%$ of cases.⁷ Lack of systematic pre-PCI imaging may have led to inadequate lesion preparation in calcified lesions; for example, nearly 50% of lesions in IVUS-CHIP were severely calcified, but advanced lesion modification was infrequent (only 6.5% atherectomy and 10.6% lithotripsy).⁵ However, pre-PCI imaging may not be feasible in all cases, and it should be acknowledged that data supporting improved outcomes with routine pre-PCI IVUS are not robust.⁵⁴

Finally, small reference vessel diameter is a major determinant of adverse outcomes after stent implantation.⁵⁵ In the current analysis, the weighted

FIGURE 6 All-Cause Mortality

(A) Direct comparison meta-analysis of ICI-guided PCI compared with angiography-guided PCI in all-cause mortality. (B) Trial sequential analysis evaluating for a 20% relative reduction with ICI-guided PCI with $\alpha = 5\%$ and power = 80%. Abbreviations as in Figure 1.

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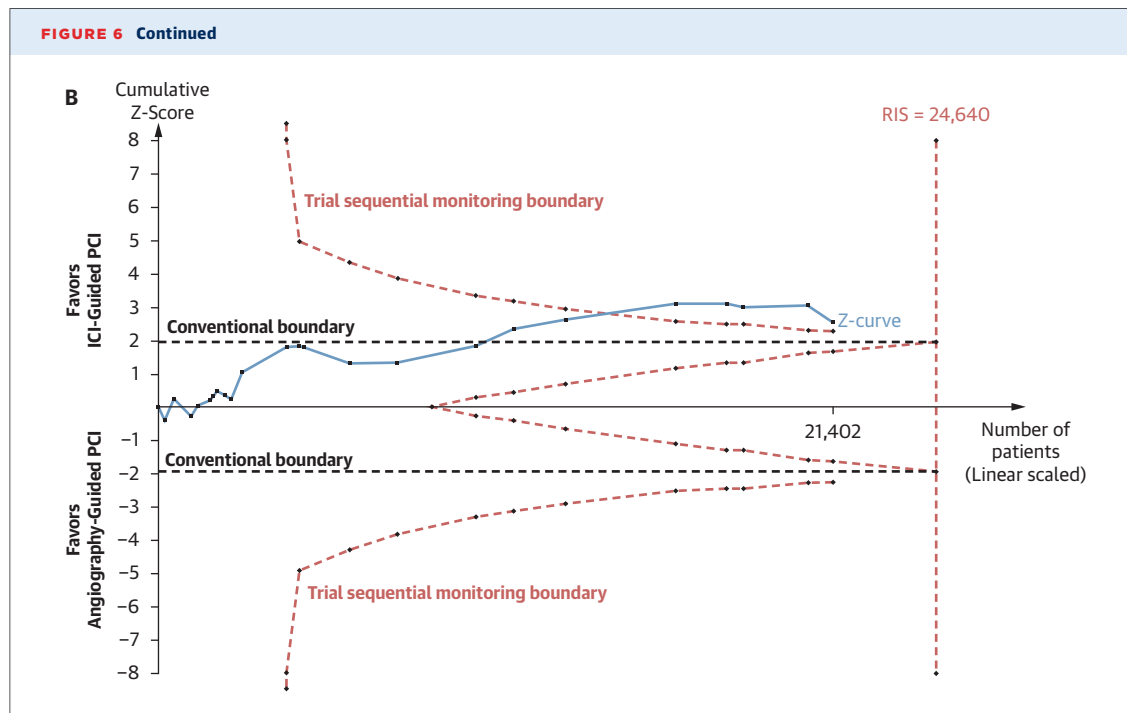
average stent diameter in the angiography guidance arm was smaller in patients from Asian trials (3.05 ± 0.11 mm) compared with those from non-Asian trials (3.26 ± 0.25 mm), supporting a potentially greater benefit of ICI-guided stenting in the smaller vessels of Asian patients.

The observation that the relationship between trial location and efficacy of ICI guidance for TLF was mitigated after adjustment for baseline lesion characteristics and procedural technique suggests that the lack of effect in some trials from Europe is not just related to geography but may be explained by the types of lesions selected or the interventions used to correct suboptimal imaging. Operator experience

may also have contributed to HTEs, as reported in prior studies.⁵⁶ Unfortunately, the current study-level meta-analysis do not permit direct assessment of these factors; such analysis would require pooled individual-patient data. Consequently, the mechanisms underlying the observed geographic variations in unadjusted analysis remain uncertain and should be considered hypothesis generating.

However, rather than diminishing the role of ICI in Europe, the fact that ICI guidance clearly improves outcomes in most studies emphasizes the necessity to apply the best evidence from ICI trials to safely achieve the largest MSA while minimizing edge dissections and residual untreated disease. These goals

FIGURE 6 Continued



are enabled with the routine use of ICI pre-PCI, during, and post-PCI; performing optimal lesion preparation, stent sizing (diameter and length), and landing zone selection; and integration of serial ICI findings to inform iterative procedural strategy to achieve optimal imaging target criteria. These findings thus support the current guideline recommendations and the growing use of ICI, with the expectation that increasing operator experience will continue to improve outcomes of ICI-guided PCI.⁵⁶

STUDY LIMITATIONS. First, absent individual-patient data, several analyses could not be performed, including time-to-event analysis, identification of clinical subgroups or anatomical lesion subtypes that might particularly benefit from ICI guidance, and evaluation of the imaging criteria most likely to optimize clinical outcomes. Second, inter-operator variability and operator expertise were not reported in the trials and were thus not accounted for in the analysis. Third, ICI targets for optimization were not prespecified nor reported consistently among the trials. As such, we used trial-level surrogates of rates of postdilatation, as well as stent length and diameter, although these parameters may not be strong proxies for stent optimization. Fourth, we were unable to tease out the effect of pre-PCI vs post-PCI imaging. Fifth, there is potential for collinearity among the trial-level covariates included in the multivariable meta-regression models. Moreover,

meta-regression analysis can be underpowered. Consequently, the findings from the multivariable meta-regression should be interpreted cautiously and considered exploratory. Sixth, although statistical heterogeneity was low for many of the outcomes, this does not rule out clinical heterogeneity in the included trials. Seventh, mortality was relatively infrequent, and the observed mortality results were relatively more fragile (fragility index of 10 and 11 for cardiac and all-cause death, respectively) than other outcomes. Although the current study does not establish the mechanism(s) through which ICI-guided PCI optimization might reduce cardiac and all-cause mortality, it is plausible that substantial reductions in stent thrombosis, MI, and restenosis may improve survival. Finally, subgroup analyses based on geography are overly simplistic and do not take into consideration between-study differences in trial design, patient, lesion and procedural characteristics, background medical therapies, operator experience, and control event rates.

CONCLUSIONS

In this comprehensive meta-analysis of randomized trials, ICI-guided PCI with both IVUS and OCT was associated with significant reductions in TLF, MI, TVR, stent thrombosis, and mortality compared with angiography-guided PCI. TSA showed firm evidence for at least a 20% relative reduction for each of these

outcomes, with the fragility index reinforcing the robustness of these findings. The Bayesian posterior probability of benefit for all endpoints >99% further supports the frequentist outcomes. However, geographic heterogeneity across the trials was present for some endpoints, explained in large part by differences in lesion characteristics and procedural technique. Greater understanding of this heterogeneity may provide new insights into the optimal use of ICI guidance to enhance event-free survival after PCI.

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PERSPECTIVES

WHAT IS NEW? In this comprehensive meta-analysis of 28 randomized trials, ICI-guided PCI with both IVUS and OCT was associated with significant reductions in TLF, MI, TVR, stent thrombosis, and mortality compared with angiography-guided PCI; trial sequential analysis showed firm evidence for at least a 20% relative reduction for each of these endpoints. The results were not fragile, with the fragility index indicating robustness of the findings. Geographic trial-level heterogeneity of results were explained in part by differences in lesion and procedural characteristics.

WHAT ARE THE CLINICAL IMPLICATIONS?

Efforts are warranted to encourage training and standardize utilization of ICI as a routine component of state-of-the-art PCI for the treatment of complex coronary artery disease.

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KEY WORDS angiography, IVUS, OCT, percutaneous coronary intervention

APPENDIX For supplemental Methods and Results, figures, and tables, please see the online version of this article.